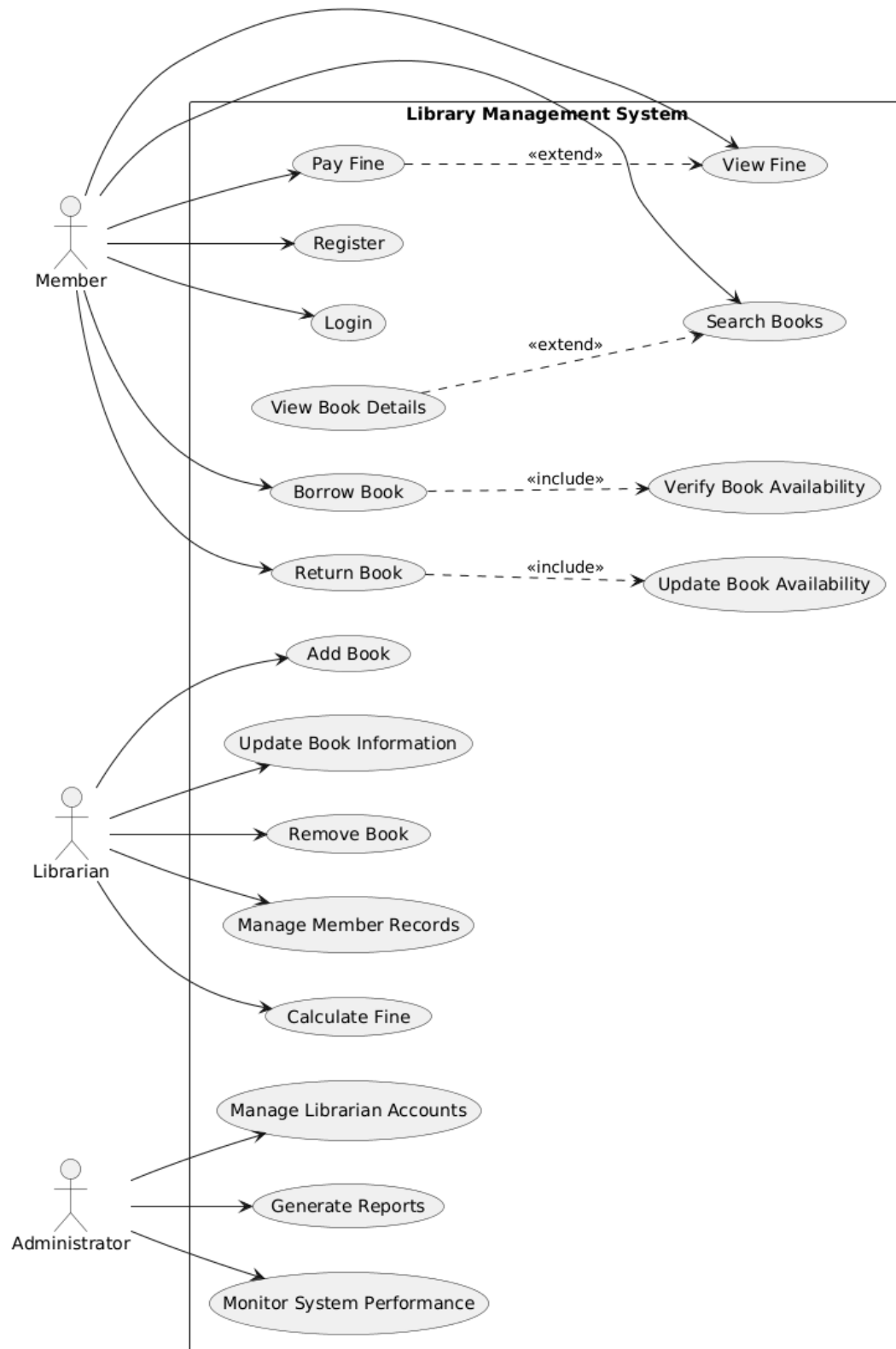


Case Study 1: Library Management System



ACTORS IDENTIFIED

1. Member
2. Librarian
3. Administrator

USE CASES IDENTIFIED

Member

- Register
- Login
- Search Books
- View Book Details
- Borrow Book
- Return Book
- View Fine
- Pay Fine

Librarian

- Add Book
- Update Book Information
- Remove Book
- Manage Member Records
- Calculate Fine

Administrator

- Manage Librarian Accounts
- Generate Reports
- Monitor System Performance

System Use Cases

- Verify Book Availability
- Update Book Availability

RELATIONSHIPS

Association

- Member ↔ Search Books
- Member ↔ Borrow Book
- Member ↔ Return Book

- Member ↔ View Fine
- Member ↔ Pay Fine
- Librarian ↔ Add Book
- Librarian ↔ Update Book Information
- Librarian ↔ Remove Book
- Librarian ↔ Manage Member Records
- Librarian ↔ Calculate Fine
- Administrator ↔ Manage Librarian Accounts
- Administrator ↔ Generate Reports
- Administrator ↔ Monitor System Performance

Include

- Borrow Book <> Verify Book Availability
- Return Book <> Update Book Availability

Extend

- View Book Details <> Search Books
- Pay Fine <> View Fine

Major Use Case 1: Borrow Book

Use Case Name: Borrow Book

Actor: Member

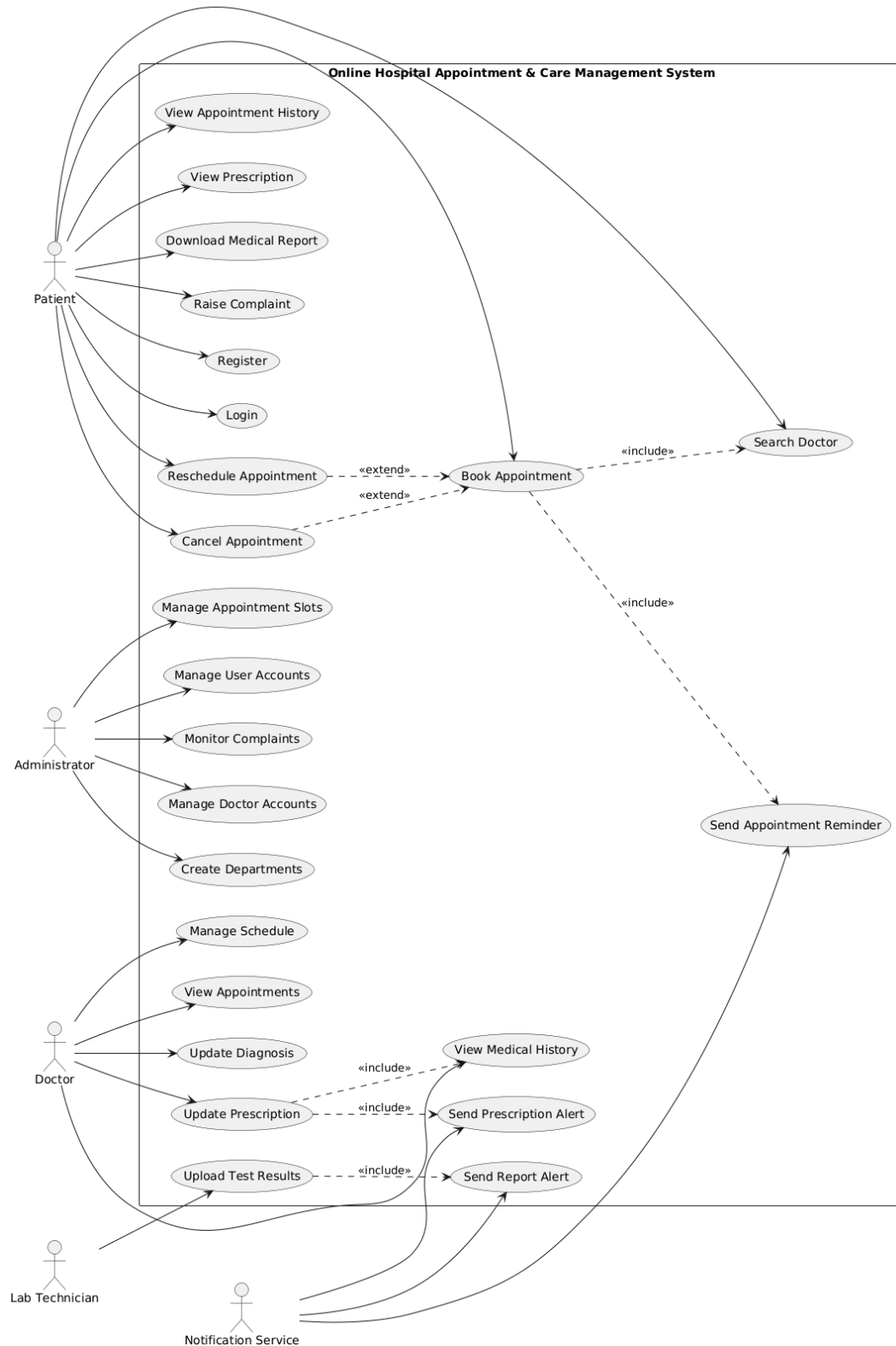
Precondition: Member is logged in.

Main Flow:

1. Member searches for a book.
2. Member selects a book.
3. System verifies availability.
4. System issues the book.
5. System updates records.
6. Borrowing is confirmed.

Postcondition: Book is successfully issued.

Case Study 2: Online Hospital Appointment & Care Management



ACTORS IDENTIFIED

1. Patient
2. Doctor
3. Lab Technician
4. Hospital Administrator
5. Notification Service (External System)

USE CASES IDENTIFIED

Patient

- Register
- Login
- Search Doctor
- Book Appointment
- Reschedule Appointment
- Cancel Appointment
- View Appointment History
- View Prescription
- Download Medical Report
- Raise Complaint

Doctor

- Manage Schedule
- View Appointments
- View Medical History
- Update Diagnosis
- Update Prescription

Lab Technician

- Upload Test Results

Hospital Administrator

- Manage Doctor Accounts
- Create Departments
- Manage Appointment Slots
- Manage User Accounts
- Monitor Complaints

Notification Service

- Send Appointment Reminder
- Send Prescription Alert
- Send Report Availability Alert

RELATIONSHIPS

Association

- Patient ↔ Book Appointment
- Patient ↔ Reschedule Appointment
- Patient ↔ Cancel Appointment
- Doctor ↔ Manage Schedule
- Doctor ↔ Update Diagnosis
- Lab Technician ↔ Upload Test Results
- Administrator ↔ Manage Doctor Accounts

Include

- Book Appointment <> Search Doctor
- View Prescription <> View Medical History

Extend

- Reschedule Appointment <> Book Appointment
- Cancel Appointment <> Book Appointment